BrightNest

Grounded after-hours help for parents

COMP6708 Mini Demo · **Group 15**

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seanfontaine.dev/COMP6708_A3_Demo

One sentence

1

A learning-centre assistant that

answers from centre truth,
acts with tools,
and hands off when humans must.

Training-free · RAG · hybrid routing · live on the public web

The problem

1

Imagine you run a tutoring centre.

- Parents message after hours: typhoons, fees, seats, trials
- Staff cannot live on WhatsApp 24/7
- A wrong fee or hours answer costs trust — fast

So you hired Group 15

We built **BrightNest**: grounded answers, structured actions, honest hand-offs.

BrightNest

Not a chatbot that invents tuition from thin air.

Technique: training-free RAG + tools

1

Why training-free? Centre docs change weekly ⇒ fine-tuning is the wrong loop.

RAG (self-built)

1. Chunk markdown KB (`content/`)
2. Embed with DashScope
3. Store / query in Chroma
4. Lexical rerank → top-k
5. Generate with Qwen + citations

Retrieval $s(q, c) = \mathrm{sim}(E(q), E(c))$ · inspector shows rank, s, chunk.

Hybrid router

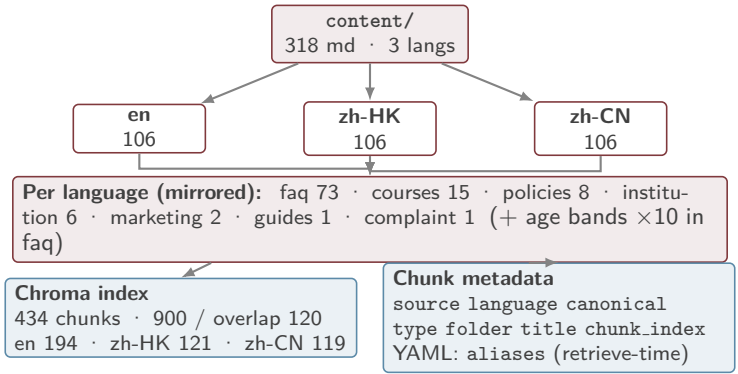
- Scores = **detector awards**, not RAG sim
- Soft FAQ: grounded compose (card > KB)
- Hard paths **bypass retrieve**
- Route arbiter if low confidence
- **Final check** before book / escalate / refuse

card privileged **KB may enrich**

proceed / clarify / abort

Knowledge base structure

1



Tool facts (typhoon signals) sit in `backend/facts/severe_weather.json`; grounded compose may still enrich from KB.

Three answer tiers

1

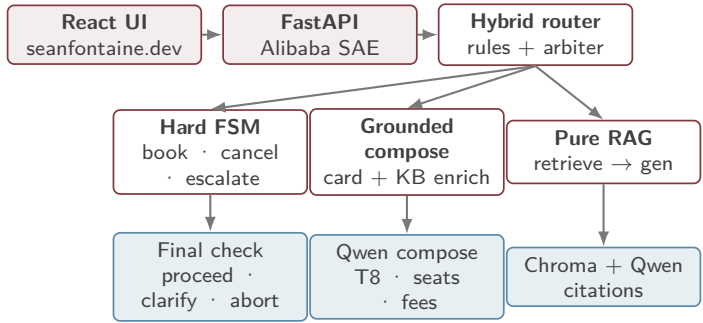
Tier	Examples	LLM / RAG
Hard FSM + final check	Book / cancel, escalate, refuse	Bypass retrieve · cheap go-ahead
Grounded compose	Typhoon+HKO, seats, fees, allergy	Live doc + KB enrich
Pure RAG	Private 1:1, multi-topic narrative	Full retrieve → generate

Intent score e.g. `fees=0.90` = fee detector fired.

Final check returns `proceed` / `clarify` / `abort` (safety fail-opens).

Architecture

1



Inspector: match cue · score ledger · final check · grounding · RAG expand

What you will see live

2

Parent chat (WhatsApp-shaped)

- Messaging surface + probes
- Chat only — no booking form

System inspector

- Router decision + score ledger
- Final check · grounding · RAG expand
- Class register · session memory

Staff desk

- SheetPro follow-ups (gaps / assessment)
- MailBox urgent inbox
- Drafted hand-off emails

5-minute rule

Slides cue. We **operate** the system.

▶ **LIVE** Open the live demo fullscreen. Reset once.

Demo beats (cue card)

2

Beat	Show	Point at
1	Typhoon	Score 1.0 · live/policy privileged
2a	Phonics fee	fees=0.90 · grounded compose
2b	RAG · Private 1:1	Retrieve → expand chunk
3	Seats → book Jason	Register · **final check** · SheetPro
4	Scuba (unknown)	Honest gap · **SheetPro** row
5	Safety allergy call	**MailBox** · safety fail-open

► **LIVE** Operate the live system. Narrate the inspector — do not read this table.

Impact & why it works

3

Impact

- 24/7 triage without inventing policy
- Tools for hours / fees / seats
- RAG for programme narrative
- Humans only when stakes demand it

Creativity

- Hybrid router + commit final check
- Staff desk = Sheets + Gmail clones

Correctness cues

Score ledger + decision path
Citations with retrieval scores
Final check before irreversible writes
Urgent MailBox + drafted email

Evaluation

Live path correctness · visible RAG · latency feel.
No formal user study required.

Thank you

3 Impact

Grounded answers. Structured actions. Honest hand-offs.

RAG · tools · staff desk

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Q&A

Thank you for listening!
Your feedback will be highly appreciated!